IV Bolus Practice 3

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

1a 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: a. Initial concentration of drug in the plasma

1b 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: b. Elimination rate constant

1c 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: c. Total body clearance

1d 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: d. Concentration of drug in the plasma 8 hours after administration of the dose

1e 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: e. Amount of drug in the body 12 hours after administration of the dose

2a 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

$A = 10.21$ mg/L, $B = 8.53$ mg/L, $\alpha = 3.14$ hr^{-1}, $\beta = 0.198$ hr^{-1}

a. What is the elimination half-life in these volunteers?

2b 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

A = 10.21 mg/L, B = 8.53 mg/L, α = 3.14 hr^{-1}, β = 0.198 hr^{-1}

b. What is the initial concentration of drug in the plasma in these volunteers?

2c 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

A = 10.21 mg/L, B = 8.53 mg/L, α = 3.14 hr^{-1}, β = 0.198 hr^{-1}

c. What is the concentration of drug in the plasma 4 hours after administration of the dose?

2d 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

A = 10.21 mg/L, B = 8.53 mg/L, $\alpha = 3.14\ hr^{-1}$, $\beta = 0.198\ hr^{-1}$

d. What is the volume of distribution of the central compartment?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a Her key

Answer: 75 mg/L

WHAT THE STEM GIVES

weight	187-pound	*not needed: the per-kilogram dose and volume cancel it*
D_0	15 mg/kg	the numerator of C_0
t½	4 hours	*not needed for C_0*
V_D	20% of body weight	0.2 L/kg, the denominator

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 20\% \text{ of body weight} = 0.2 \text{ L/kg}$$

A volume given as a percentage of body weight is read as litres per kilogram, taking 1 kg as 1 L: 20% is 0.2 L/kg.

SETUP

$$C_0 = \frac{D_0}{V_D}$$

Right after an IV bolus the whole dose is in the body, so the concentration at time zero is the dose divided by the volume of distribution.

ALGEBRA

$$C_0 = \frac{15 \text{ mg/kg}}{0.2 \text{ L/kg}} = 75 \text{ mg/L}$$

Dose and volume are both per kilogram, so the kilograms cancel and mg/L is left; the 187-pound weight is not needed for C_0.

> **DOES THE NUMBER MAKE SENSE?**
>
> Each kilogram holds 15 mg of drug in 0.2 L, so each litre holds five times 15 mg, 75 mg; the weight cancels and does not change C_0.

1b Her key

Answer: 0.1733 hr^{-1}

WHAT THE STEM GIVES

weight	187-pound	*not needed for k*
D_0	15 mg/kg	*not needed: k does not depend on the dose*
t½	4 hours	the denominator of $\frac{0.693}{t½}$
V_D	20% of body weight	*not needed for k*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{t_{½}}$$

$t_{½} = \frac{0.693}{k}$ rearranged for k: the first-order rate constant is 0.693 divided by the half-life.

ALGEBRA

$$k = \frac{0.693}{4\ \text{hr}} = 0.1733\ \text{hr}^{-1}$$

0.693 is a pure number, so dividing by hours leaves reciprocal hours, the unit of a first-order rate constant. Her key gives k to four decimal places.

> **DOES THE NUMBER MAKE SENSE?**
> 0.693 over 4 is a little under a fifth, so k near 0.17 hr^{-1} fits, and 0.1733 hr^{-1} does. A half-life in hours gives a rate constant in hr^{-1}.

1c Her key

Answer: 2.95 L/hr

WHAT THE STEM GIVES

weight	187-pound	$\frac{187}{2.2} = 85$ kg
D_0	15 mg/kg	*not needed: clearance does not depend on the dose*
$t_{½}$	4 hours	gives $k = \frac{0.693}{4} = 0.1733\ hr^{-1}$
V_D	20% of body weight	0.2 L/kg times 85 kg gives 17 L

THE WORKING, ONE LINE AT A TIME

UNIT

$$W = \frac{187\ \text{lb}}{2.2\ \text{lb/kg}} = 85\ \text{kg}$$

Pounds to kilograms: dividing by 2.2 lb per kg cancels the pounds and leaves kilograms.

ALGEBRA

$$V_D = (0.2\ \text{L/kg})(85\ \text{kg}) = 17\ \text{L}$$

20% of body weight is 0.2 L per kg; multiplying by the weight cancels the kilograms and leaves an absolute volume in litres.

SETUP

$$Cl_T = k \times V_D, \text{ with } k = \frac{0.693}{4\ \text{hr}} = 0.1733\ \text{hr}^{-1}$$

Clearance is the elimination rate constant times the volume of distribution, so both are needed in absolute units.

ALGEBRA

$Cl_T = (17\ L)(0.1733\ hr^{-1}) = 2.95\ L/hr$

L × hr^{-1} = L/hr, a volume of plasma cleared of drug each hour. Her key does not show the 17 L; it is 85 kg × 0.2 L/kg.

> **DOES THE NUMBER MAKE SENSE?**
> About a sixth of the 17 L volume is cleared each hour (k = 0.1733 hr^{-1}), a little under 3 L/hr; 2.95 L/hr fits.

1d Her key

Answer: 18.75 mg/L

WHAT THE STEM GIVES

weight	187-pound	*not needed: C_0 is per kilogram over per kilogram*
D_0	15 mg/kg	gave C_0 in part a
t½	4 hours	k = 0.1733 hr^{-1}; 8 hours is two half-lives
V_D	20% of body weight	gave C_0 in part a
t	8 hours	the exponent kt
C_0	75 mg/L	multiplied by e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_0\, e^{-kt}$, with $k = \dfrac{0.693}{4\ hr} = 0.1733\ hr^{-1}$

After an IV bolus the concentration falls first order from C_0, so C_0 is multiplied by e^{-kt}, the fraction left at time t.

ALGEBRA

$C8 = (75\ mg/L)\, e^{-(0.1733)(8)} = (75\ mg/L)(0.25) = 18.75\ mg/L$

kt = (0.1733 hr^{-1})(8 hr) = 1.386: the hours cancel, so the exponent is a pure number, and $e^{-1.386} = 0.25$ is the fraction of C_0 left.

ALGEBRA

8 hr is 2 half-lives: 25% remains, (0.25)(75 mg/L) = 18.75 mg/L

Her key also counts half-lives: two halvings leave a quarter of the 75 mg/L, the same 18.75 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is two half-lives of 4 hours, so the 75 mg/L halves twice and a quarter of it, 18.75 mg/L, remains.

1e Her key

Answer: 159.35 mg

WHAT THE STEM GIVES

weight	187-pound	$\frac{187}{2.2} = 85$ kg
D_0	15 mg/kg	times 85 kg gives the whole dose, 1275 mg
t½	4 hours	k = 0.1733 hr^{-1}; 12 hours is three half-lives
V_D	20% of body weight	*not needed: an amount, not a concentration*
t	12 hours	the exponent kt

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (15 \text{ mg/kg})(\frac{187 \text{ lb}}{2.2 \text{ lb/kg}}) = (15 \text{ mg/kg})(85 \text{ kg}) = 1275 \text{ mg}$$

The dose per kilogram times the weight in kilograms: the kilograms cancel and the whole dose in mg is left.

SETUP

$$D_B = D_0\, e^{-kt}, \text{ with } k = \frac{0.693}{4 \text{ hr}} = 0.1733 \text{ hr}^{-1}$$

The amount in the body falls first order, like the concentration, so the dose is multiplied by the fraction left at time t.

ALGEBRA

$$D12 = (1275 \text{ mg})\, e^{-(0.1733)(12)} = 159.35 \text{ mg}$$

kt = (0.1733 hr^{-1})(12 hr) = 2.08 has no units, and $e^{-2.08}$ = 0.125 of the dose remains, so the answer keeps the unit of the dose, mg.

ALGEBRA

12 hr is 3 half-lives: 12.5% remains, (0.125)(1275 mg) = 159.375 mg

Her key also counts half-lives: three halvings leave one eighth of the dose. Both routes give about 159.4 mg.

> **DOES THE NUMBER MAKE SENSE?**
> Twelve hours is three half-lives of 4 hours, so one eighth of the 1275 mg dose remains, about 159 mg.

Her key gives 159.375 mg by counting three half-lives and 159.35 mg from the exponential with k = 0.1733 hr^{-1}; both are accepted.

2a Her key

Answer: 3.5 hr

WHAT THE STEM GIVES

D_0	300-mg	*not needed*
A	10.21 mg/L	*not needed*
B	8.53 mg/L	*not needed*
α	3.14 hr^{-1}	*not needed: the distribution phase*
β	0.198 hr^{-1}	the denominator of $\frac{0.693}{\beta}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{½\beta} = \frac{0.693}{\beta}$$

β, the smaller of the two exponents, is the slope of the terminal phase, which is elimination; the elimination half-life comes from β alone.

ALGEBRA

$$t_{½\beta} = \frac{0.693}{0.198\ hr^{-1}} = 3.5\ hr$$

A pure number over hr^{-1} leaves hours. $\alpha = 3.14\ hr^{-1}$ belongs to the fast distribution phase and is not used.

> **DOES THE NUMBER MAKE SENSE?**
> $\beta = 0.198\ hr^{-1}$ is about a fifth, so the half-life is about 0.693 × 5, 3.5 hr; a half-life from $\alpha = 3.14\ hr^{-1}$ would be a fraction of an hour.

2b Her key

Answer: 18.74 mg/L

WHAT THE STEM GIVES

D_0	300-mg	*not needed: the intercepts already include the dose*
A	10.21 mg/L	added
B	8.53 mg/L	added
α	3.14 hr^{-1}	*not needed: the distribution phase*
β	0.198 hr^{-1}	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = A + B$$

At t = 0 both exponentials in $C_p = A\,e^{-\alpha t} + B\,e^{-\beta t}$ equal 1, so the initial concentration is the sum of the two intercepts.

ALGEBRA

$$C_0 = (10.21 + 8.53)\ mg/L = 18.74\ mg/L$$

A and B are both in mg/L, so they add directly and the sum is a concentration in mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 is the sum of the two intercepts, so it is larger than either: 10.21 + 8.53 = 18.74 mg/L.

2c Her key

Answer: 3.86 mg/L

WHAT THE STEM GIVES

D_0	300-mg	*not needed*

A	10.21 mg/L	the distribution term, nearly gone by 4 hours
B	8.53 mg/L	the elimination term, almost all of C_p at 4 hours
α	3.14 hr^{-1}	$\alpha t = 12.56$, so $e^{-\alpha t}$ is negligible
β	0.198 hr^{-1}	$\beta t = 0.792$
t	4 hours	the time in both exponents

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = A\, e^{-\alpha t} + B\, e^{-\beta t}$$

The two-compartment equation: each intercept falls with its own rate constant, and the concentration is the sum of the two terms.

ALGEBRA

$$C4 = (10.21\ mg/L)\, e^{-(3.14)(4)} + (8.53\ mg/L)\, e^{-(0.198)(4)}$$

Each exponent is hr^{-1} times hr, a pure number: 12.56 for the distribution term and 0.792 for the elimination term.

ALGEBRA

$$C4 = 0.00004\ mg/L + (8.53\ mg/L)(0.4529) = 3.86\ mg/L$$

$e^{-12.56}$ is so small that the A term adds almost nothing; by 4 hours distribution is over and only the β term remains.

> **DOES THE NUMBER MAKE SENSE?**
>
> By 4 hours the α term, with α = 3.14 hr^{-1}, has fallen to almost nothing, so C_p is about B $e^{-\beta t}$: 8.53 mg/L times a little under a half, 3.86 mg/L.

2d Her key

Answer: 16 L

WHAT THE STEM GIVES

D_0	300-mg	the numerator
A	10.21 mg/L	part of A + B = 18.74 mg/L
B	8.53 mg/L	part of A + B
α	3.14 hr^{-1}	*not needed: the distribution phase*
β	0.198 hr^{-1}	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$V_c = \frac{D_0}{A + B}$$

At time zero the whole dose is in the central compartment at the concentration A + B, so the dose over that concentration is the central volume.

ALGEBRA

$$A + B = (10.21 + 8.53)\ \text{mg/L} = 18.74\ \text{mg/L}, \text{ so } V_C = \frac{300\ \text{mg}}{18.74\ \text{mg/L}} = 16\ \text{L}$$

mg over mg/L leaves litres: $\frac{300}{18.74} = 16.01$ L, which her key reports as 16 L.

DOES THE NUMBER MAKE SENSE?

300 mg over about 19 mg/L is a little over 15 L; 16 L fits, and it must be smaller than the 300 L that 1 mg/L would give.